Knowledge Organiser
Year 7: Forces – Balanced Forces

What happens when a force acts on an object?
- When a force acts on an object it can either make it change shape, direction or speed
- The size of the force impacts the magnitude of the change
- Usually more than one force is acting on an object, so we must determine the resultant force



Kicking a football makes it accelerate



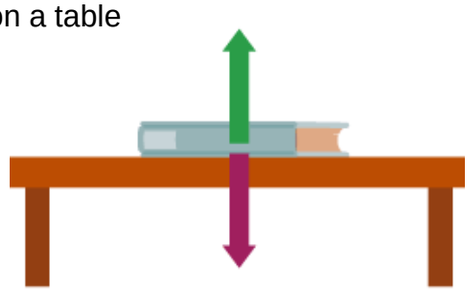
Squeezing a sponge make it change shape



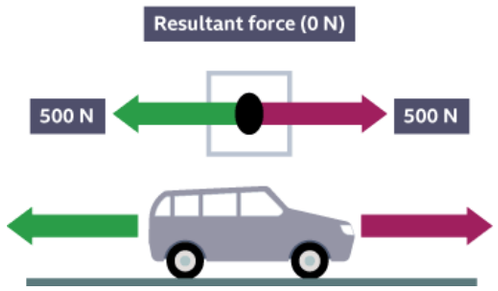
Turning a steering wheel makes the wheels change direction

How do we draw forces?
- We can't see forces but can see their effects
- We draw forces as **arrows** on **force diagrams**
- The arrow shows the direction the force is acting
- The **size of the arrow** shows the size of the force

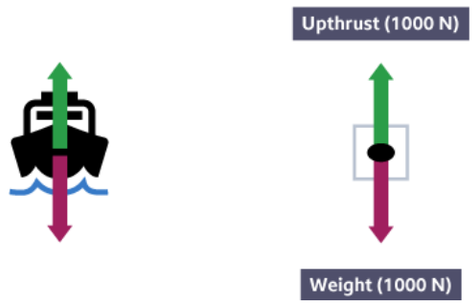
A force diagram showing the forces acting on a book resting on a table



What are balanced forces?
- If the forces acting on an object are balanced we say there is no **resultant force**
- This means that the forces are the same size but acting in opposite directions
- If forces on an object are balanced, it is either **stationary** or **travelling at a constant speed**

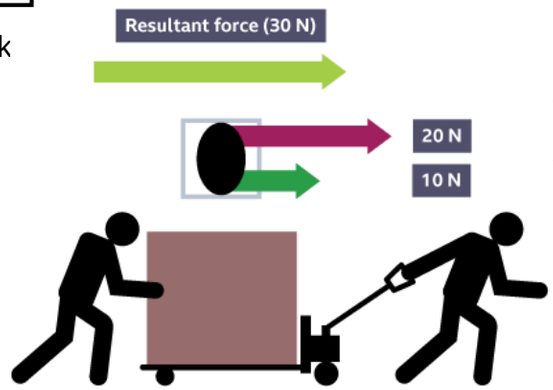


A force diagram showing the forces acting on a car travelling at a constant speed

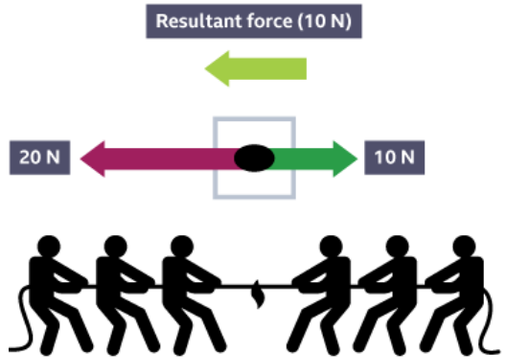


A force diagram showing the forces acting on a boat floating on a still ocean

What is a resultant force?
- When multiple forces act on an object at once, the resultant force is the overall force that acts on the object
- Forces are **vectors,** this means the direction they act is important

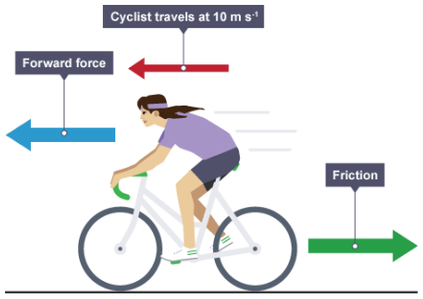


A force diagram showing the result of two forces of different sizes acting in the same direction

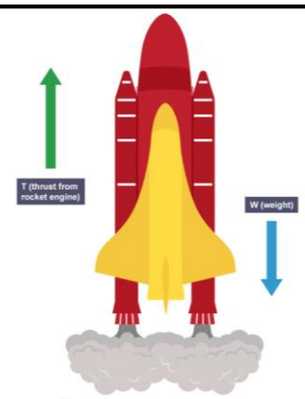


A force diagram showing the result of two forces of different sizes acting in the opposite directions

What if the forces are unbalanced?
- When forces are unbalanced it causes the speed of an object to change
- This can be **acceleration,** where the object gains speed
- Or **deceleration,** where the object loses speed



A force diagram showing the a cyclist travelling at a constant speed due to the balanced forces acting on them



A force diagram showing a rocket that is accelerating upwards due to the unbalanced forces acting on it

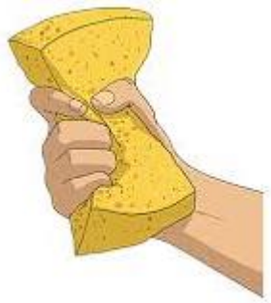
RAG Yourself!

Learning Outcomes;	RAG
State what is meant by a resultant forces	
Calculate resultant forces on an object	
Describe what is meant by balanced forces	
Draw force diagrams using arrows	
Explain why an object will start to accelerate	
Explain how an object travels at a constant speed	

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A force diagram showing the forces acting on a book resting on a table

What are balanced forces?

A force diagram showing the forces acting on a car travelling at a constant speed

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What is a resultant force?

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What if the forces are unbalanced?

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